

# Specification of Python §1—2026 edition

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The language “Python § $x$ ” is a sublanguage of [Python 3.13](#) and defined in the documents titled “Python § $x$ ”, where  $x$  refers to the respective textbook chapter.

## 1 Syntax

A Python program is a *program*, defined using Backus-Naur Form<sup>1</sup> as follows:

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<sup>1</sup>We adopt Henry Ledgard’s BNF variant that he described in *A human engineered variant of BNF*, ACM SIGPLAN Notices, Volume 15 Issue 10, October 1980, Pages 57-62. In our grammars, we use **bold** font for keywords, *italics* for syntactic variables,  $\epsilon$  for nothing,  $x \mid y$  for  $x$  or  $y$ ,  $[x]$  for an optional  $x$ ,  $x\dots$  for zero or more repetitions of  $x$ , and  $(x)$  for clarifying the structure of BNF expressions.

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>program</i>         | ::= <i>import-stmt... block</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | program                                                                                                                                                                                                                                                                                                                                                                                          |
| <i>import-stmt</i>     | ::= <b>from</b> <i>dotted-name</i> <b>import</b> <i>import-clause</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | import statement                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>dotted-name</i>     | ::= <i>name</i> ( <i>.</i> <i>name</i> )...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | dotted identifier chain                                                                                                                                                                                                                                                                                                                                                                          |
| <i>import-clause</i>   | ::= <i>import-as-names</i><br>  ( <i>import-as-names</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | wildcard or name list                                                                                                                                                                                                                                                                                                                                                                            |
| <i>import-as-names</i> | ::= <i>import-as-name</i> ( , <i>import-as-name</i> )...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | clause name list                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>import-as-name</i>  | ::= <i>name</i> [ <b>as</b> <i>name</i> ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | clause name and alias                                                                                                                                                                                                                                                                                                                                                                            |
| <i>statement</i>       | ::= <i>name</i> = <i>expression</i><br>  <b>def</b> <i>name</i> ( <i>names</i> ) : <i>block</i><br>  <b>return</b> <i>expression</i><br>  <i>if-statement</i><br>  <i>expression</i>                                                                                                                                                                                                                                                                                                                                                                      | variable declaration<br>function declaration<br>return statement<br>conditional statement<br>expression statement                                                                                                                                                                                                                                                                                |
| <i>names</i>           | ::= $\epsilon$   <i>name</i> ( , <i>name</i> )...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | name list                                                                                                                                                                                                                                                                                                                                                                                        |
| <i>if-statement</i>    | ::= <b>if</b> <i>expression</i> : <i>block</i><br>( <b>elif</b> <i>expression</i> : <i>block</i> )...<br><b>else</b> : <i>block</i>                                                                                                                                                                                                                                                                                                                                                                                                                       | conditional statement                                                                                                                                                                                                                                                                                                                                                                            |
| <i>block</i>           | ::= <i>statement...</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | block statement                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>expression</i>      | ::= <i>integer</i><br>  <i>float</i><br>  <i>complex</i><br>  <b>True</b>   <b>False</b><br>  <b>None</b><br>  <i>string</i><br>  <i>name</i><br>  <i>expression</i> <i>binary-operator</i> <i>expression</i><br>  <i>unary-operator</i> <i>expression</i><br>  <i>expression</i> <i>binary-logical</i> <i>expression</i><br>  <i>expression</i> ( <i>expressions</i> )<br>  <b>lambda</b> ( <i>name</i>   <i>names</i> ) : <i>expression</i><br>  <i>expression</i> <b>if</b> <i>expression</i> <b>else</b> <i>expression</i><br>  ( <i>expression</i> ) | primitive integer expression<br>primitive float expression<br>primitive complex expression<br>primitive boolean expression<br>primitive None expression<br>primitive string expression<br>name expression<br>binary operator combination<br>unary operator combination<br>logical composition<br>function application<br>lambda expression<br>conditional expression<br>parenthesised expression |
| <i>binary-operator</i> | ::= <b>+</b>   <b>-</b>   <b>*</b>   <b>/</b>   <b>**</b>   <b>%</b><br>  <b>==</b>   <b>!=</b>   <b>&gt;</b>   <b>&lt;</b>   <b>&gt;=</b>   <b>&lt;=</b>                                                                                                                                                                                                                                                                                                                                                                                                 | binary operator                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>unary-operator</i>  | ::= <b>not</b>   <b>-</b>   <b>+</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | unary operator                                                                                                                                                                                                                                                                                                                                                                                   |
| <i>binary-logical</i>  | ::= <b>and</b>   <b>or</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | logical composition symbol                                                                                                                                                                                                                                                                                                                                                                       |
| <i>expressions</i>     | ::= $\epsilon$   <i>expression</i> ( , <i>expression</i> )...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | argument expressions                                                                                                                                                                                                                                                                                                                                                                             |

## Indentation

In Python, [indentation](#) is syntactically significant and strictly enforces code block structure. Unlike languages using braces, Python employs whitespace (spaces or tabs) to delimit blocks for control flow (e.g., `if`, `for`, `while`), function definitions, and class declarations. Key rules:

- **Consistency:** Use 4 spaces per indentation level (PEP 8 recommendation). Mixing tabs and spaces is prohibited.
- **Alignment:** Statements within the same block must align vertically. Misaligned indentation raises an `IndentationError`.
- **Nesting:** Each nested block increases indentation by one level. Dedenting resumes the outer block.
- **Line Continuation:** For multi-line statements, align wrapped lines with the opening delimiter or use an extra level.

## Restrictions

- In Python, a logical line is the smallest unit of code that the interpreter can execute, typically corresponding to a complete statement. A physical line, by contrast, is a single line in the source file, usually ending with a newline character. A logical line may span multiple physical lines depending on whether line joining is used.

By default, each physical line corresponds to a logical line. However, when a statement is too long, Python allows it to be split across multiple physical lines using line continuation. Python provides two ways to join lines: **explicit line joining** and **implicit line joining**.

**Explicit line joining** uses a backslash (`\`) to join the current physical line with the next one, forming a single logical line. The backslash must be the last character on the line—no characters (including whitespace or comments) are allowed after it.

```
total = 1 + 2 + 3 + \
        4 + 5
```

**Implicit line joining** occurs when expressions are enclosed in parentheses `()`, square brackets `[]`, or braces `{}`. Within these delimiters, line breaks are allowed without using a backslash.

```
total = (
    1 + 2 + 3 +
    4 + 5
)
```

Therefore, when breaking a statement across lines, you must either use a backslash for explicit continuation or wrap the expression in parentheses, brackets, or braces for implicit continuation. Unlike JavaScript, which uses automatic semicolon insertion, Python requires each line to be syntactically complete—it does not infer the end of a statement implicitly.

For details on logical lines, see the official Python documentation: [Python Lexical Analysis – Logical Lines](#).

- Return statements are only allowed in bodies of functions. Each return statement must appear on a single logical line.
- Lambda expressions are limited to a single logical line.
- Re-declaration variables or functions is not allowed. Once a variable or function is defined, it cannot be redefined with the same name in the same scope <sup>2</sup>.

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<sup>2</sup>Scope refers to the region of a program in which a particular name (such as a variable, function, or class) is defined and can be accessed. In other words, it determines the part of the program where you can use that name without causing a name error. Scope is determined by the program's structure (usually its lexical or textual layout) and governs the visibility and lifetime of variables and other identifiers.

In [Python](#), the *scope* of a declaration is determined lexically: a variable declared inside a function is local to that function; if it is declared outside any function, it is global (i.e., module-level). Moreover, if a variable is declared in an enclosing function, it is available to inner functions (the enclosing scope), and if not found in these scopes, Python looks into the built-in scope.

Reassignment is not allowed. Within any scope, each name (including function parameters, variables, and function names) must be declared or defined at most once. Therefore, a parameter or declared variable must not appear on the left-hand side of an assignment, and a name must not be defined by multiple assignments or function definitions in the same scope.

## Names

Names<sup>3</sup> start with `_`, or a letter<sup>4</sup> and contain only `_`, letters or digits<sup>5</sup>. Reserved words<sup>6</sup> are not allowed as names.

Valid names are `x`, `_45`, and `π`, but always keep in mind that programming is communicating and that the familiarity of the audience with the characters used in names is an important aspect of program readability.

## Numbers

Python supports three numeric types: integers (`int`), floats (`float`), and complex numbers (`complex`).

### Integers

Integers can be represented in decimal notation, optionally prefixed with a sign (+ or -). Additional base notations are supported, such as binary (`0b1010` or `0B1010`), octal (`0o777` or `0O777`), and hexadecimal (`0x1A3F` or `0X1A3F`). Underscores (`_`) may be used for readability (e.g., `1_000_000`). Examples include `42`, `-0b1101`, and `0xFF_00`.

### Floats

Floats use decimal notation with an optional decimal dot. Scientific notation (multiplying the number by  $10^x$ ) is indicated with the letter `e` or `E`, followed by the exponent `x`. Special values `inf` (infinity), `-inf`, and `nan` (not a number) are allowed. Examples include `3.14`, `-0.001e+05`, and `6.022E23`.

### Complex Numbers

Complex numbers are written as `<real>±<imag>j`, where `j` (or `J`) denotes the imaginary unit. Both the real and imaginary parts are stored as floats. The imaginary part is mandatory (e.g., `5j`, `0j`, and `0+3j` is valid, `5` alone is real). Examples include `2+3j`, `-4.5J`, `0j`, and `1e3-6.2E2J`.

## Strings

Strings are of the form `"double-quote-characters"`, where *double-quote-characters* is a possibly empty sequence of characters without the character `"` and without the newline character; of the form `'single-quote-characters'`, where *single-quote-characters* is a possibly empty sequence of characters without the character `'` and without the newline character; and of the form `'''triple-single-quote-characters'''`, or `"""triple-double-quote-characters"""`, where *triple-double-quote-characters* is a possibly empty sequence of characters and can span multiple lines and may contain both single and double quotes without escaping.

The following characters can be represented in strings<sup>7</sup> as given:

- `\<newline>`: Backslash followed by a newline is ignored.

<sup>3</sup>In [Python 3.13 Documentation](#), these names are called *identifiers*.

<sup>4</sup>By *letter* we mean [Unicode](#) letters (L) or letter numbers (NI).

<sup>5</sup>By *digit* we mean characters in the [Unicode](#) categories Nd (including the decimal digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9), Mn, Mc and Pc.

<sup>6</sup>By *reserved word* we mean any of: `False`, `await`, `else`, `import`, `pass`, `None`, `break`, `except`, `in`, `raise`, `True`, `class`, `finally`, `is`, `return`, `and`, `continue`, `for`, `lambda`, `try`, `as`, `def`, `from`, `nonlocal`, `while`, `assert`, `del`, `global`, `not`, `with`, `async`, `elif`, `if`, `or`, `yield`. These are all reserved words, or keywords of the language that cannot be used as ordinary identifiers.

<sup>7</sup>In [Python 3.13 Documentation](#), unless an `'r'` or `'R'` prefix is present, escape sequences in string and bytes literals are interpreted according to rules similar to those used by Standard C. *identifiers*.

- `\\`: Represents a backslash (i.e., `\`).
- `\'`: Represents a single quote (`'`).
- `\"`: Represents a double quote (`"`).
- `\a`: Represents the ASCII Bell (BEL) character.
- `\b`: Represents the ASCII Backspace (BS) character.
- `\f`: Represents the ASCII Formfeed (FF) character.
- `\n`: Represents the ASCII Linefeed (LF) character.
- `\r`: Represents the ASCII Carriage Return (CR) character.
- `\t`: Represents the ASCII Horizontal Tab (TAB) character.
- `\v`: Represents the ASCII Vertical Tab (VT) character.

## 2 Dynamic Type Checking

Expressions evaluate to integers, floats, complex numbers, boolean values, strings or function values. Implementations of Source generated error messages when unexpected values are used as follows.

Only function values can be applied using the syntax:

$$expression ::= name( expressions )$$

For compound functions, implementations need to check that the number of *expressions* matches the number of parameters.

The following table specifies what arguments Python's operators take and what results they return. Implementations need to check the types of arguments and generate an error message when the types do not match.

| operator                              | argument 1          | argument 2          | result  |
|---------------------------------------|---------------------|---------------------|---------|
| <code>+, -, *, %</code>               | int                 | int                 | int     |
| <code>/</code>                        | int                 | int                 | float   |
| <code>+, -, *, /, %</code>            | int                 | float               | float   |
| <code>+, -, *, /, %</code>            | float               | int, float          | float   |
| <code>+, -, *, /</code>               | int                 | complex             | complex |
| <code>+, -, *, /</code>               | float               | complex             | complex |
| <code>+, -, *, /</code>               | complex             | int, float, complex | complex |
| <code>+</code>                        | string              | string              | string  |
| <code>**</code>                       | int                 | int $\geq$ 0        | int     |
| <code>**</code>                       | int                 | int $<$ 0           | float   |
| <code>**</code>                       | int                 | float               | float   |
| <code>**</code>                       | int                 | complex             | complex |
| <code>**</code>                       | float               | int, float          | float   |
| <code>**</code>                       | float               | complex             | complex |
| <code>**</code>                       | complex             | int, float, complex | complex |
| <code>==, !=</code>                   | int, float, complex | int, float, complex | bool    |
| <code>==, !=</code>                   | string              | string              | bool    |
| <code>&gt;, &gt;=, &lt;, &lt;=</code> | int, float          | int, float          | bool    |
| <code>&gt;, &gt;=, &lt;, &lt;=</code> | string              | string              | bool    |
| <code>and</code>                      | bool                | any                 | any     |
| <code>or</code>                       | bool                | any                 | any     |
| <code>not</code>                      | bool                |                     | bool    |
| <code>-</code>                        | int                 |                     | int     |
| <code>-</code>                        | float               |                     | float   |
| <code>-</code>                        | complex             |                     | complex |

Following `if` and `elif`, Python §*x* only allows boolean expressions. Lambda expressions can only be used as arguments to operators when they are parenthesized.

### 3 Standard Library

The standard library contains constants and functions that are always available in this language. In py-slang, the standard library functions are not implemented using a conventional split between primitive and predeclared functions. Unlike in Source, where predeclared functions are defined using built-in primitives, all standard library functions in py-slang are written directly in TypeScript and embedded into the runtime of the CSE Machine. These functions are treated as part of the host environment rather than as user-level definitions, and they interact directly with the evaluator's internal control stack and environment. This design simplifies the interpreter architecture and allows for tighter integration with the execution model, enabling better support for visualization, error handling, and platform-specific extensions within the Source Academy learning environment.

The standard library for Python §*x* is documented in [online documentation](#) of Python §*x*.

#### MISC Library

The following names are provided by the MISC library:

- `complex(x), str(x)`: *primitive*, converts the value *x* to a complex number, or a string representation of *x* respectively, and returns the respective value.
- `complex(r, i)`: *primitive*, takes in two numbers *r* and *i*, and returns the complex number with real part *r* and imaginary part *i*.
- `complex()`: *primitive*, returns the complex number with real part 0 and imaginary part 0.

- `repr(s)`: *primitive*, returns a string representation of `s` that is unambiguous and suitable for debugging; for many values, this string can be used to reconstruct `s`.
- `is_boolean(x)`, `is_integer(x)`, `is_float(x)`, `is_string(x)`, `is_none(x)`, `is_function(x)`, `is_complex(x)`: *primitive*, returns `True` if the type of `x` matches the function name and `False` if it does not.
- `real(x)`, `imag(x)`: *primitive*, return the real and imaginary parts of a complex number `x` respectively.
- `input(s)`: *primitive*, pops up a window that displays the *string* `s`, provides an input line for the user to enter a text, a “Cancel” button and an “OK” button. The call of `input` suspends execution of the program until one of the two buttons is pressed. If the “OK” button is pressed, `input` returns the entered text as a string. If the “Cancel” button is pressed, `input` returns a non-string value.
- `min(x, y, *args)` and `max(x, y, *args)`: *primitive*, takes in two or more arguments and returns the smallest and largest of the arguments respectively. If multiple arguments are equally small or large, the first one encountered in the argument list is returned. The arguments must be mutually comparable (i.e., they must support the `<` and `>` operators).
- `abs(x)`: *primitive*, returns the absolute value of `x`. For an `int` input, it returns the non-negative integer equivalent. For a `float` input, it returns the positive floating-point number representing its magnitude. For a `complex` input, it returns the modulus of the complex number, which is a non-negative float.
- `print(*x)`: *primitive*, takes in any number of parameters, converts them to their string representations using `str` and writes them to the console, followed by a newline character.
- `error(*x)`: *primitive*, takes in any number of parameters, converts them to their string representations using `str` and writes them to the console, followed by a newline character. The evaluation of any call of `error` aborts the running program immediately.
- `arity(f)`: *primitive*, takes in a function `f` and returns the minimum number of parameters that `f` can accept.
- `round(x)`<sup>8</sup>: *primitive*, takes in an `int` or `float` `x` and returns the nearest integer to `x`.
- `round(x, n)`: *primitive*, takes in an `int` or `float` `x` and an `int` `n`, and returns the value of `x` rounded to `n` digits of precision after the decimal point. If the second argument `n` is `None`, it is treated as `round(x)`.
- `len(s)`: *primitive*, takes in a container for a finite number of values (e.g., a string), and returns the number of elements in the container.
- `time_time()`: *primitive*, returns number of milliseconds elapsed since January 1, 1970 00:00:00 UTC
- `random_random()`: *primitive*, returns a random float uniformly in the range `[0.0, 1.0)`.

All library functions can be assumed to run in  $O(1)$  time, except `print`, `error`, `str` and `repr`, which run in  $O(n)$  time, where  $n$  is the size (number of components such as pairs) of their arguments.

## Deviations from native Python

We intend the Python §*x* to be a conservative extension of native Python: Every correct Python §*x* program should behave *exactly* the same using a Python §*x* implementation, as it does using a native Python implementation. We assume, of course, that suitable libraries are used by the TypeScript implementation, to account for the predefined names of Python §*x*. This section lists some exceptions where we think a Python §*x* implementation should be allowed to deviate from the native Python specification, for the sake of internal consistency and esthetics.

<sup>8</sup>In Python 3.13, when the value to be rounded is exactly halfway between two possible rounded values, it rounds to the nearest even number. For example, `round(2.5)` would return 2, while `round(3.5)` would return 4.

**Output Behavior Differences Between py-slang and Standard Python REPL:** Python provides an interactive mode, where users can type and immediately evaluate Python expressions line by line. This mode is entered when you simply type `python` in a terminal without specifying a file. This is commonly referred to as the Python REPL. In the standard Python REPL, any evaluated expression automatically has its result printed to the console, even if the user does not explicitly call `print()`. For example:

```
>>> 1 + 2
3
>>> "hello"
'hello'
```

This is because Python's REPL implicitly displays the return value of each expression, unless it is `None`.

In contrast, `py-slang` implements the behavior of non-interactive Python. Only expressions explicitly passed to `print()` produce output. If the expression is evaluated without a `print` call, no output will appear, even though a value is computed.

For example, in `py-slang`:

```
print(1 + 2)      # Outputs: 3
```

But

```
1 + 2            # Outputs nothing
```